ECONOMETRICS PROJECT · GROUP 5 · TÉLÉCOM PARIS

Do higher minimum wages reduce in-work poverty?

Evidence from 21 EU countries, 2009 to 2024

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The EU expects minimum wages to reduce in-work poverty

DIRECTIVE (EU) 2022/2041

Adequate minimum wages

One aim: reduce in-work poverty (Art. 5(1))

Reference values: 60% of the median wage or 50% of the average wage (Art. 5(4))

EU SIMULATION (JRC, 2021)

−12 to −13%

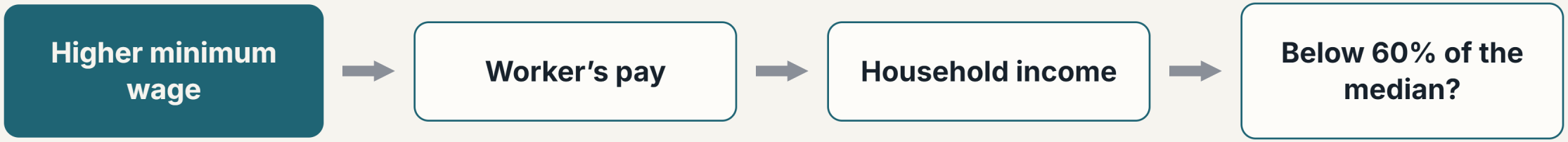
in-work poverty, if every minimum wage reached these reference values

Our question: in the real data, were years with a higher minimum wage also years with less in-work poverty?

Sources: Directive (EU) 2022/2041; Grünberger et al. (2021), JRC Working Paper 06/2021

In-work poverty depends on the whole household

In-work poor: worked at least half of the year, and household income (adjusted for size) is below 60% of the national median.



WHAT CAN BREAK THE LINK

Hours
Part-time work, or work for only part of the year

Household
A partner's income lifts it; children need more

Benefits
Some benefits fall when pay rises

The line moves
60% of a median that can rise too

Definition: Eurostat, EU-SILC methodology. Studies: Dube (2019); Backhaus & Müller (2023); Giupponi et al. (2024)

21 countries, 16 years, all from Eurostat

330 country-years
income years 2009 to 2024

BE	BG	CZ	DE	EE	EL	ES
FR	HR	HU	IE	LT	LU	LV
MT	NL	PL	PT	RO	SI	SK

Every EU country with a national statutory minimum wage. Germany from 2015.

The survey asks about last year's income, so each rate is matched with the minimum wage of the year the income was earned.

Outcome: in-work poverty of employees
Aged 18 to 64. Average 6.3%.

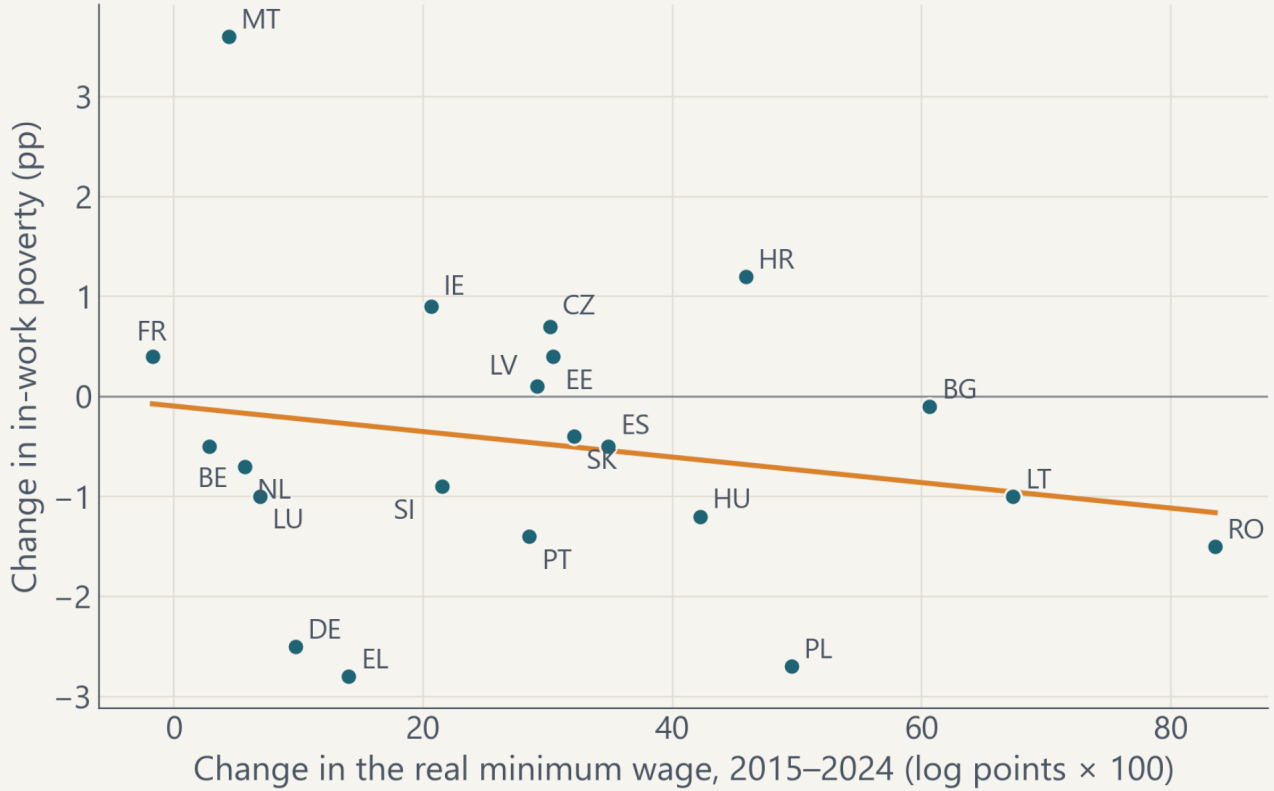
Real minimum wage
Monthly minimum wage divided by the country's price index

Kaitz index
Minimum wage as a % of the average wage

Controls
Unemployment rate, GDP per person, part-time share

Source: Eurostat database (EU-SILC, minimum wages, prices, labour force survey), downloaded on 7 November 2026

Bigger minimum wage increases did not bring bigger falls in poverty



SLOPE OF THE LINE

−0.013

pp per log point (robust SE 0.013)

Each dot is one country, changes from 2015 to 2024.

France: almost no change. Romania: real minimum wage more than doubled.

In-work poverty of employees, income years 2015 and 2024. Source: Eurostat

We compare each country with itself over time

$$IWP_{it} = \beta\, MW_{it} + \gamma' X_{it} + \alpha_i + \delta_t + u_{it}$$

α_i Country dummies

Remove what is permanent: benefit system, family structure

δ_t Year dummies

Remove shocks common to all: COVID, energy prices in 2022

X Controls

Unemployment rate, GDP per person, part-time share

β / 10 = change in pp when the minimum wage is 10% higher

Errors clustered by country, plus **wild cluster bootstrap** p-values

MW = log real minimum wage or Kaitz index. One country and one year dummy dropped (dummy variable trap).

Main result: a higher real minimum wage goes with almost no change

Real minimum wage +10%

−0.03 pp

in-work poverty of employees (average 6.3%)

95% interval: **−0.29 to +0.24**

Bootstrap p = 0.85

Rules out falls bigger than about 5% of the average rate

Same answer in every specification

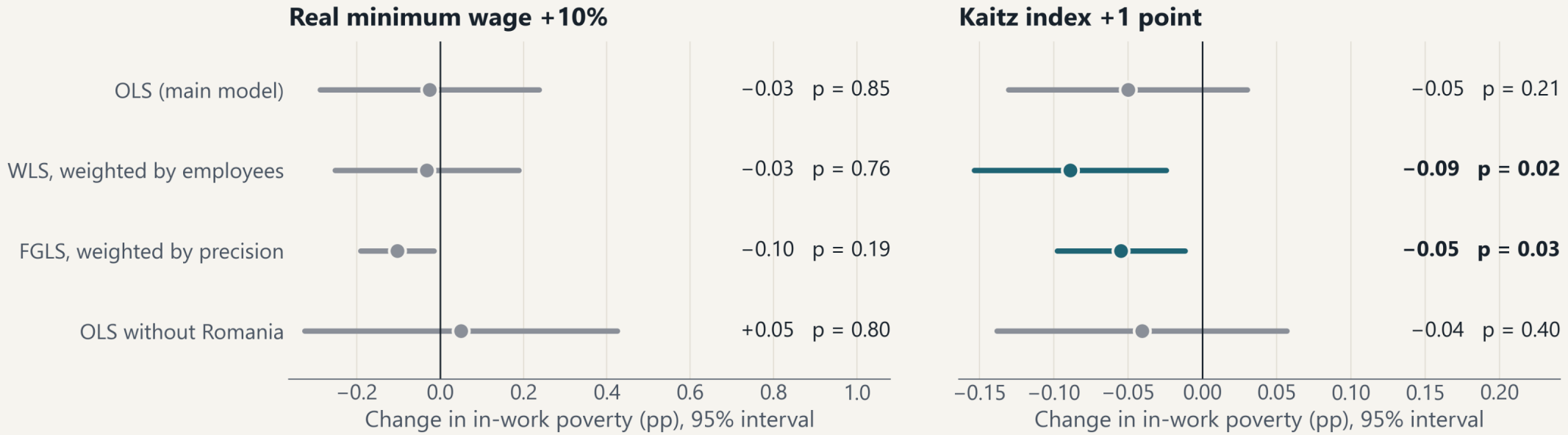
Model	per +10%
Pooled OLS	−0.05
Country dummies + controls	−0.03
Country + year dummies	−0.01
Country + year dummies + controls	−0.03

None is significant: every bootstrap p is above 0.7.

Kaitz index +1 point: **−0.05 pp**
95% interval −0.13 to +0.03, bootstrap p = 0.21

330 country-years, 21 countries. Standard errors clustered by country, t with 20 degrees of freedom.

Weighting the countries: a small Kaitz effect appears



WLS

Weight = number of employees. Describes the average employee instead of the average country.

FGLS

Weight = 1 / residual variance of the country. Countries with precise data count more.

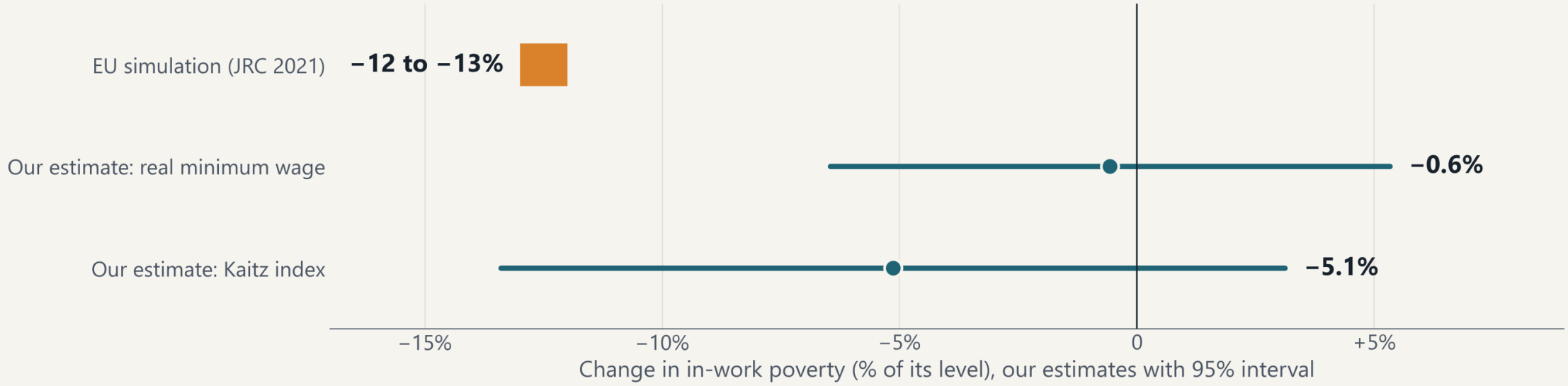
Few clusters, again

FGLS, real minimum wage: clustered p = 0.02, bootstrap p = 0.19. Four countries hold 55% of the weight.

Teal = bootstrap p below 0.05. p = wild cluster bootstrap p-value. Lines: 95% intervals from clustered standard errors.

Our estimates are much smaller than the EU simulation

Scenario: every minimum wage rises to 50% of the average wage. In 2019, 17 of 20 countries were below this level. They needed **+16%** on average.



The simulation keeps behaviour fixed and covers all workers. Our estimates include whatever happened after real increases.

Simulation: Grünberger et al. (2021), JRC, EUROMOD. Ours: employees, two-way FE with controls.

The result survives many robustness checks

No controls −0.01	**First differences** −0.04	**Without the COVID years** −0.03	**Dummies for series breaks** +0.13
Survey-year timing −0.05	**Prices of the poorest 20%** −0.02	**Without Romania** +0.05	**One country dropped** −0.12 to +0.05

Lead test
Next year's minimum wage does not predict poverty today: −0.18, p = 0.34

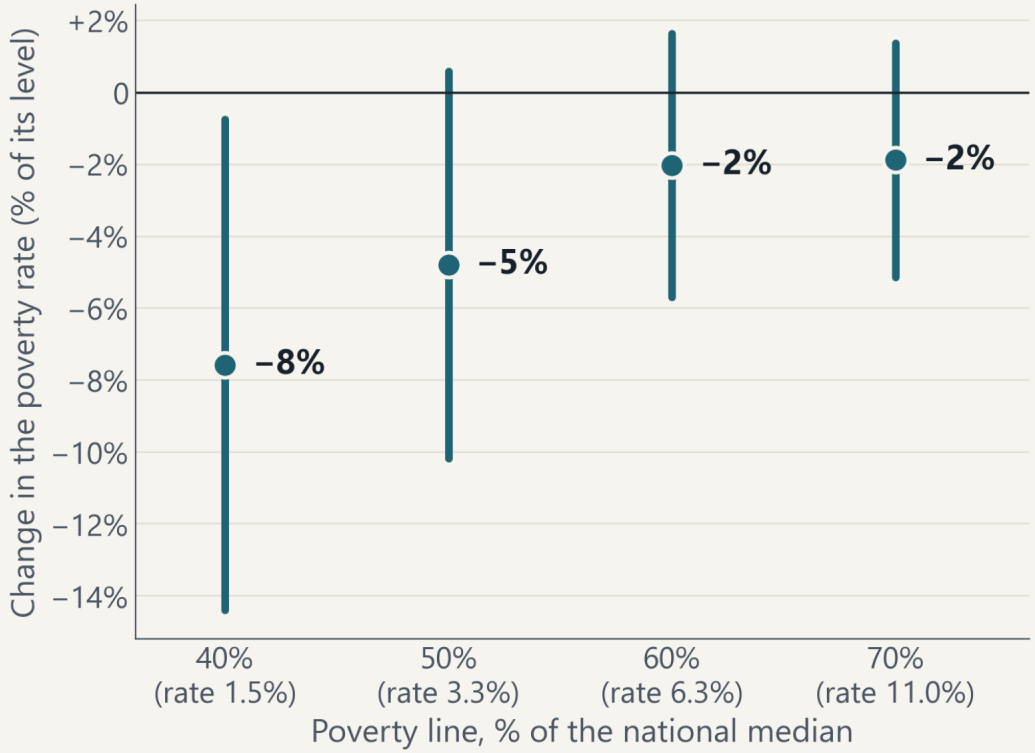
Placebo: retired people
Their poverty does not move with the minimum wage: +0.04 (SE 0.39)

Selection out of work
Employment of low-educated adults does not fall: −0.03 (SE 0.60)

Real minimum wage +10%, change in pp. No estimate is significant: every bootstrap p-value is above 0.5.

Why so weak? Three hints from the data

1 Deep poverty responds more



2 The poverty line moves too

Minimum wage +10% goes with a real poverty line about **3.4%** higher (imprecise)

3 The household matters

In-work poverty, pp per +10% minimum wage

Single adult	−0.74
Single adult with children	−0.67
Couple with children	−0.19
Couple without children	0.00

Same model, other outcomes. None is significant at 5%. Household types: all employed (no split for employees).

What our results cannot tell us

Associations only

Minimum wages respond to the economy and to politics, and taxes and benefits change at the same time.

Only country averages

We do not see who earns the minimum wage, or in which household.

Gross pay only

Romania 2018: the gross increase compensated workers for social contributions moved from employers to them.

Few countries

21 clusters give wide intervals, and two-way FE can weight different effects in odd ways.

Romania: Eurofound (Fric, 2018). Two-way FE weights: de Chaisemartin and D'Haultfœuille (2020).

CONCLUSION

A higher minimum wage alone did little against in-work poverty

Real minimum wage +10%

−0.03 pp

About zero (interval −0.29 to +0.24)

Kaitz index +1 point

−0.05 pp

Small, significant only with weights

Minimum wage at 50% of the average

−0.6%

The EU simulation said −12 to −13%

In-work poverty also depends on hours, the household and benefits.

Thank you

Questions are welcome

One command, **python run_all.py**, rebuilds every number, table and figure from our Eurostat snapshot.

Code, data, notebook and report: github.com/VictorZaca/minimum-wages-in-work-poverty-eu

BACKUP

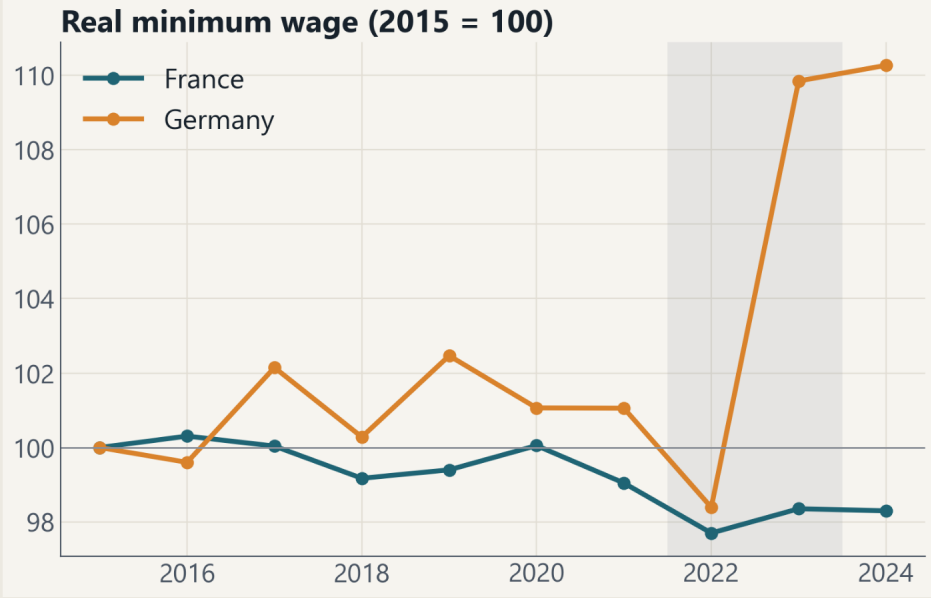
Diagnostics and other specifications

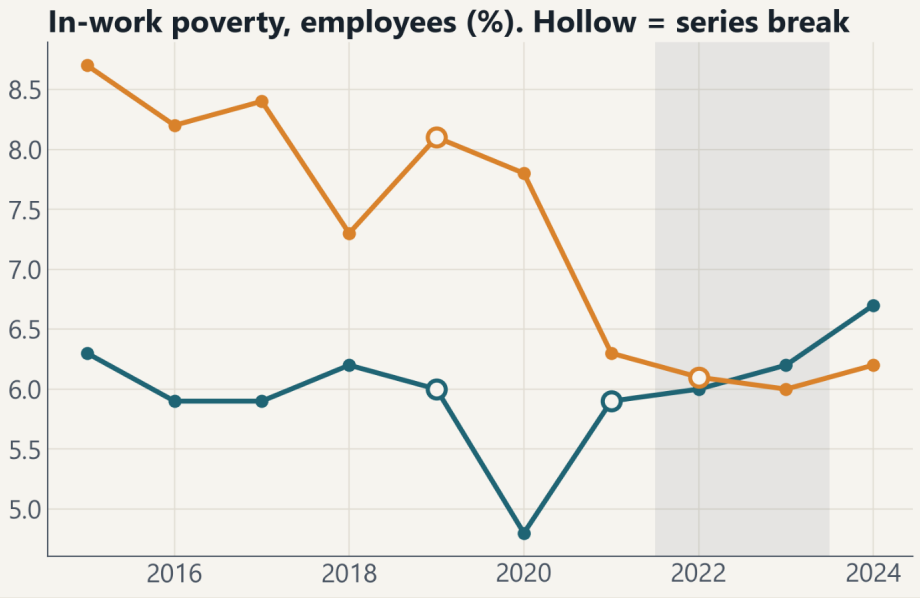
Test or specification (main model)	Result
Breusch-Pagan test (39 df)	LM = 95.3, p < 0.001
Special White test (2 df)	LM = 16.2, p < 0.001
Residual SD by country	0.35 (BE, NL) to 1.8 (HU)
Residual autocorrelation, AR(1)	ρ = 0.55
Controls, joint F test	p = 0.86
Year dummies, joint F test	p = 0.19 (0.05 without controls)
Within R^2	0.005
Minimum detectable effect, 80% power	0.37 pp per +10%
Outcome in logs	elasticity −0.11 (SE 0.18)
Quadratic in the Kaitz index	joint F test p = 0.22
Standardized (within-country SDs)	−0.05 (real MW), −0.15 (Kaitz)

Two-way FE with controls, 330 country-years. F tests use clustered standard errors.

BACKUP

France and Germany: two countries are not enough





France: real minimum wage always within 2.3% of its 2015 value. Poverty 6.3% → 6.7%.

Germany: +10% by 2024, almost all after October 2022. Poverty 8.7% → 6.2%, but mostly in 2019 to 2021, before the increase.

Grey band: inflation surge of 2022 and 2023. Hollow markers: breaks in the EU-SILC series (Germany 2019 and 2022).

BACKUP

Why employees? In Romania, poverty fell outside wage employment

All workers, 2015 → 2024

18.6% → 10.7%

Employees, 2015 → 2024

5.8% → 4.3%

Real minimum wage

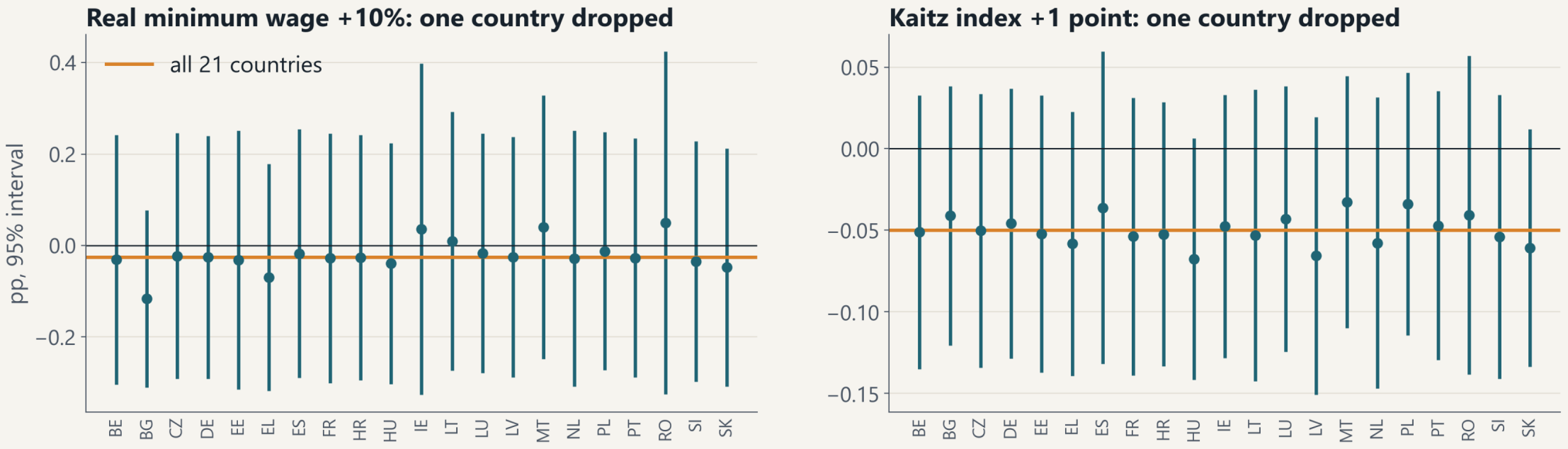
+131%

Estimate (pp)	All 21 countries	Without Romania
All workers, real minimum wage +10%	−0.16	+0.12
All workers, Kaitz +1 point	−0.13	−0.05
Employees, real minimum wage +10%	−0.03	+0.05
Employees, Kaitz +1 point	−0.05	−0.04

Gap between the two poverty rates: 11 points in Romania, 1.3 points in the median other country.

BACKUP

Dropping one country at a time changes little



Real minimum wage: **−0.12 to +0.05** per +10%. Kaitz index: **−0.07 to −0.03** per point. Largest Cook's distance: 0.06 (Lithuania, 2009).

Orange line: estimate with all 21 countries. Lines: 95% intervals, clustered standard errors.

BACKUP

What moves the Kaitz index?

Kaitz = minimum wage ÷ average wage. It rises when the minimum wage rises, and also when average pay falls. So we split it into its two parts.

Change per +10% (SE)	Real minimum wage	Real average wage
In-work poverty, employees (pp)	−0.18 (0.16)	+0.19 (0.24)
Deprivation, poorest 20% (pp)	+1.6 (0.9)	−3.3 (0.8)

A higher Kaitz index goes with **more** deprivation (+0.49, p = 0.02). The split shows it comes from the average wage: bad years have low pay and high deprivation. A change in the Kaitz index is not always a change in policy.

Deprivation: material and social deprivation rate of the bottom income quintile, dated by survey year.